scientific reports

Check for updates

OPEN

Green spectrofluorimetric determination of Ibrexafungerp using NBD-Cl derivatization

Omkulthom Al kamaly[1] & Mustafa Sayedahmed[2]✉

Ibrexafungerp (IBX) is the first approved triterpenoid antifungal indicated for the treatment of vulvovaginal candidiasis, particularly in cases resistant to azole and echinocandin therapies. In this study, a sensitive, simple, and environmentally sustainable spectrofluorimetric method was developed for the quantitative determination of IBX in pharmaceutical tablets. The method is based on chemical derivatization of IBX with 4-chloro-7-nitrobenzo-2-oxa-1,3-diazole (NBD-Cl), forming a highly fluorescent adduct measured at an emission wavelength of 554 nm following excitation at 472 nm. The reaction proceeds via nucleophilic substitution between the primary amine group of IBX and NBD-Cl. Experimental conditions affecting derivatization and fluorescence intensity were systematically optimized. The method exhibited excellent linearity over the concentration range of 150–2000 ng/mL (r^2 = 0.9996), with limits of detection and quantification of 39.196 and 118.774 ng/mL, respectively. Validation according to ICH guidelines confirmed satisfactory accuracy, precision, and robustness. The method was successfully applied without interference from excipients. Greenness assessment using Analytical Eco-Scale, AGREE, and GAPI tools confirmed the environmentally friendly nature of the proposed method, supporting its suitability for routine quality control applications.

Keywords Ibrexafungerp, Chemical derivatization, NBD-Cl, Green spectrofluorimetric method

Ibrexafungerp (IBX), Fig. 1, is a novel triterpenoid antifungal agent that exhibits potent activity against a broad spectrum of fungal pathogens, including azole and echinocandin-resistant strains[1,2]. It acts by inhibiting β-(1,3)-D-glucan synthase, an essential enzyme for fungal cell wall synthesis, leading to cell lysis and death[3,4]. Due to its unique mechanism of action and oral bioavailability, IBX has emerged as an important therapeutic option for the treatment of vulvovaginal candidiasis and other invasive fungal infections[5–7].

Despite its clinical importance, the development of simple, sensitive, and environmentally friendly analytical methods for IBX determination remains limited. Reported methods, particularly reversed-phase high-performance liquid chromatography (RP-HPLC)[8], provide acceptable accuracy and precision; however, they are associated with several drawbacks. These include relatively narrow linearity ranges at higher concentration levels (15–90 µg/mL), lower sensitivity (limit of detection = 0.13 µg/mL), and dependence on organic solvents such as acetonitrile, which increases both cost and environmental impact. Additionally, these methods often require complex sample preparation and longer analysis times.

Spectrofluorimetry offers a valuable analytical alternative due to its inherent high sensitivity, simplicity of instrumentation, and reduced solvent usage[9]. Compared with conventional spectrophotometric and chromatographic techniques, fluorescence-based methods can achieve lower detection limits and faster analysis, making them particularly attractive for routine quality control and bioanalytical applications[10]. However, the direct fluorometric determination of IBX is hindered by its structural characteristics, as the molecule lacks strong native fluorophores and exhibits limited conjugation, resulting in weak intrinsic fluorescence.

To overcome this limitation, derivatization strategies can be employed to introduce fluorescent moieties into the analyte structure[11]. Among the various fluorogenic reagents, 4-chloro-7-nitrobenzo-2-oxa-1,3-diazole (NBD-Cl) is widely utilized due to its high reactivity toward nucleophilic functional groups, particularly primary and secondary amines[12,13]. Given that IBX contains an accessible aliphatic primary amine, it is well-suited for derivatization with NBD-Cl via nucleophilic substitution, leading to the formation of a highly fluorescent product.

The present study was undertaken to develop a sensitive, reliable, and environmentally sustainable spectrofluorimetric method for the quantitative determination of IBX in pharmaceutical dosage forms. Given

[1]Department of Pharmaceutical Sciences, College of Pharmacy, Princess Nourah bint Abdulrahman University, P.O. Box 84428, Riyadh 11671, Saudi Arabia. [2]Analytical Research and Development Department, Benchmark Health Company, Cairo, Egypt. ✉email: mustafasayedahmed14@gmail.com

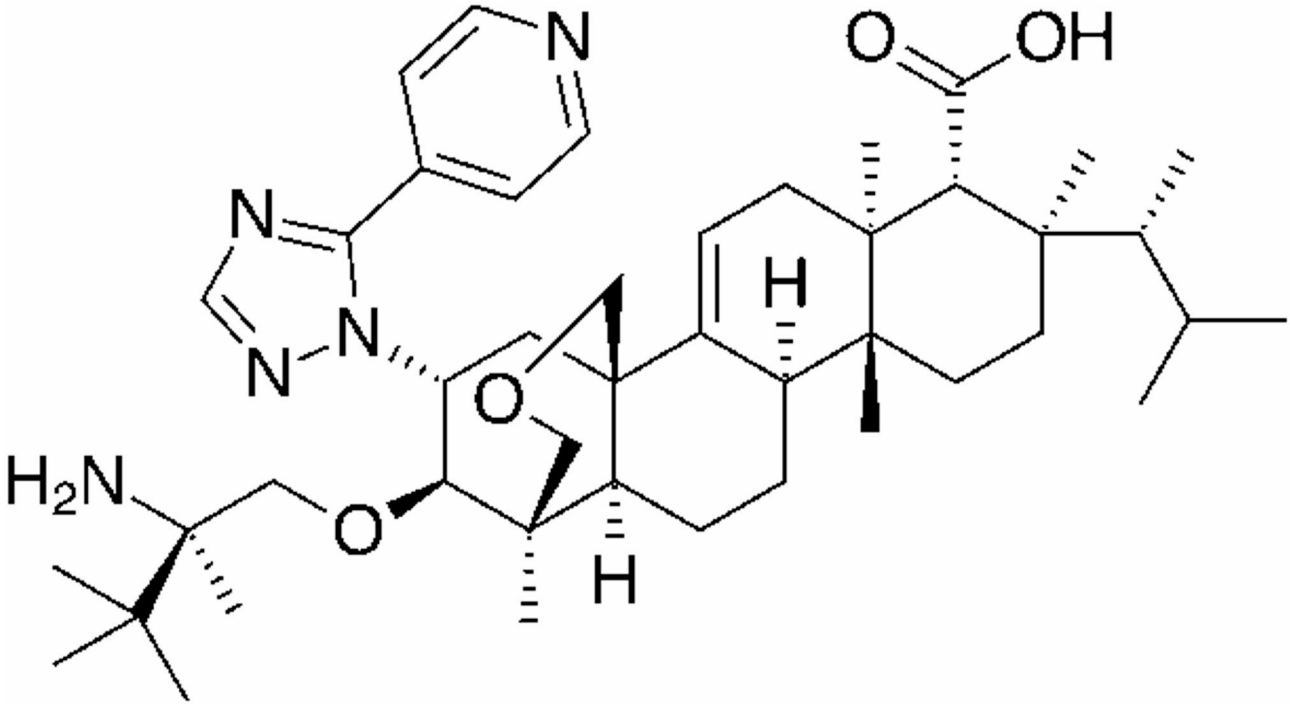


Fig. 1. Chemical structure of IBX.

the absence of strong native fluorescence in IBX, the method is based on a derivatization approach utilizing NBD-Cl as a fluorogenic reagent to generate a highly fluorescent derivative amenable to spectrofluorimetric detection.

A key objective of this work is to systematically optimize the experimental variables governing the derivatization reaction, including reagent concentration, pH, temperature, reaction time, and diluting solvent, in order to maximize fluorescence intensity and analytical sensitivity. Furthermore, the proposed method aims to provide a practical alternative to chromatographic techniques by offering simplicity, reduced cost, and simpler operational requirements without compromising accuracy and precision. In addition, the method is intended to be validated in accordance with international guidelines to ensure its suitability for routine quality control applications. Finally, in line with current trends in green analytical chemistry, this work seeks to evaluate the environmental impact of the developed method using multiple greenness assessment tools, with the goal of establishing a balance between analytical performance and ecological sustainability.

Experimental

Materials

Ibrexafungerp (IBX) reference standard (purity 99.38%, batch no. 724715), 4-chloro-7-nitrobenzo-2-oxa-1,3-diazole (NBD-Cl), Dimethyl sulfoxide (DMSO), methanol, ethanol, and acetonitrile were kindly supplied by Sigma-Aldrich (Germany). Commercial tablets (Brexafemme®, labeled to contain 150 mg IBX per tablet) were obtained from SCYNEXIS (New Jersey, USA). Analytical-grade hydrochloric acid, boric acid, potassium chloride, and sodium hydroxide chemicals were obtained from El-Nasr Company, Egypt. Buffer solutions were prepared according to the specifications of the United States Pharmacopoeia (USP).

Instrument

Fluorescence measurements were performed using a Jasco FP-6200 spectrofluorometer (Japan) equipped with a 150 W xenon lamp as the excitation source. Measurements were carried out using 1 cm quartz cells, with both excitation and emission slit widths set at 10 nm. The scan speed was set to medium.

Preparation of solutions

A standard stock solution of IBX (100 µg/mL) was prepared by accurately weighing 10 mg of IBX, dissolving in approximately 60 mL DMSO, and completing to volume with the same solvent in a 100-mL volumetric flask. Working standard solutions (1.5–20 µg/mL) were obtained by appropriate dilution of the stock solution with DMSO. The NBD-Cl solution (0.2% w/v in methanol) was freshly prepared daily, protected from light, and stored at low temperature (4 °C). Under these conditions, the solution remained stable for up to 24 h.

General fluorescence procedure

Aliquots of 1.0 mL of IBX working solutions were transferred into a series of 10-mL stoppered test tubes. To each tube, 1.5 mL of NBD-Cl solution and 1.0 mL of borate buffer (pH 8.5) were added. The reaction mixtures were

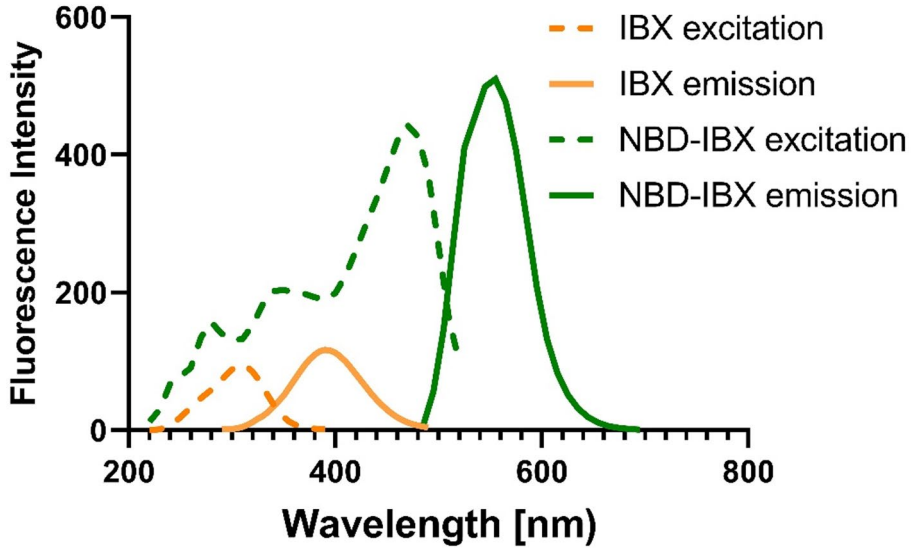


Fig. 2. Fluorescence spectra of the IBX alone (10 µg/mL) and the NBD-IBX derivative at (1.4 µg/mL).

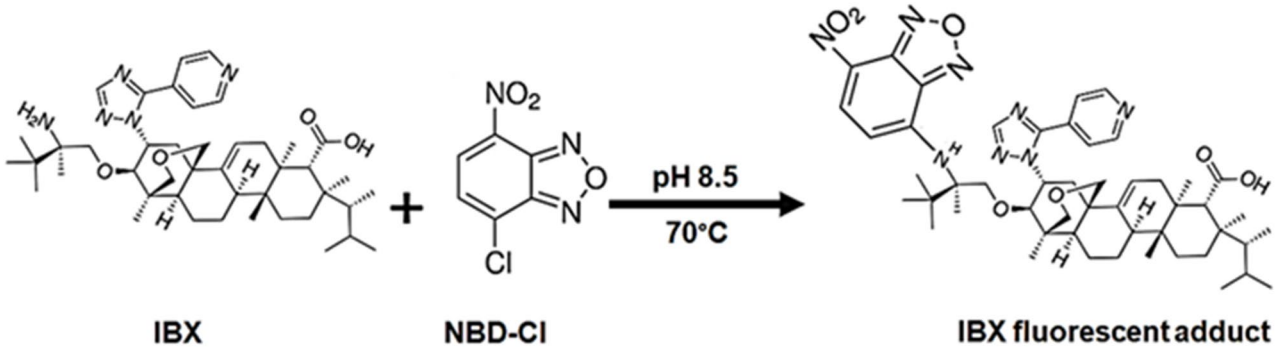


Fig. 3. Suggested reaction pathway of IBX with NBD-Cl.

heated in a thermostatically controlled water bath at 70 °C for 25 min to allow complete derivatization. After cooling rapidly in an ice bath, 1.0 mL of 1 M hydrochloric acid was added. The solutions were quantitatively transferred into 10-mL volumetric flasks and diluted to volume with DMSO. Fluorescence intensity was measured at an emission wavelength of 554 nm after excitation at 472 nm. A reagent blank was prepared similarly. Calibration curves were constructed by plotting corrected fluorescence intensity (sample minus blank) versus final IBX concentration (150–2000 ng/mL).

Analysis of the tablets

Four Brexafemme® tablets were accurately weighed and finely powdered. An amount of powder equivalent to 10 mg IBX was transferred into a 100-mL volumetric flask, extracted with 60 mL DMSO, and sonicated for 15 min to ensure complete dissolution. The solution was filtered through quantitative filter paper (Whatman No. 1, 11 µm pore size), and the volume was completed with DMSO. Appropriate dilutions were made to obtain concentrations within the calibration range, and the general fluorescence procedure was applied.

Results and discussion

Method development and optimization

Under alkaline conditions, IBX undergoes derivatization with NBD-Cl, yielding a highly fluorescent yellow product. The resulting derivative exhibited a maximum emission at 554 nm upon excitation at 472 nm (Fig. 2). This reaction is attributed to a nucleophilic substitution mechanism, whereby the chlorine atom of NBD-Cl is displaced by the nucleophilic amino group present in IBX, forming a stable fluorescent adduct (Fig. 3). This structural modification enhances the fluorometric response and enables sensitive detection of the drug.

To achieve optimal analytical performance, the experimental variables influencing the derivatization reaction were systematically investigated while maintaining a constant IBX concentration. All experiments were performed in triplicate, and the results are presented as mean ± relative standard deviation (RSD). The effect of pH was first examined, considering that the NBD-Cl reaction with amines is favored under alkaline conditions[13]. Borate

buffer systems covering the pH range 7.0–10.0 were evaluated. Maximum fluorescence intensity was observed at pH 8.5, which was therefore selected as the optimal reaction medium (Fig. 4). In addition, the influence of buffer volume was assessed, and 1.0 mL of borate buffer (pH 8.5) was found to provide the highest and most consistent fluorescence response. It is well established that NBD-Cl is susceptible to hydrolysis in alkaline media, leading to the formation of NBD-OH, a fluorescent by-product that may contribute to background signal. To minimize this interference, the reaction mixture was acidified after completion of the derivatization step. The addition of 1.0 mL of 1 M hydrochloric acid effectively suppressed the fluorescence of NBD-OH without adversely affecting the emission intensity of the IBX derivative, thereby improving method selectivity.

The influence of NBD-Cl concentration was also investigated by varying the volume of a 0.2% (w/v) solution. The fluorescence intensity increased with increasing reagent volume up to 1.25 mL, beyond which no significant enhancement was observed. Accordingly, 1.5 mL was selected as the optimum volume. Temperature and reaction time are critical parameters for ensuring complete derivatization. The reaction mixtures were heated at different temperatures and for varied durations. Maximum fluorescence intensity was achieved upon heating at 70 °C for 25 min, indicating complete reaction under these conditions (Fig. 4). Prolonged heating or higher temperatures did not yield further improvement and were therefore avoided.

Finally, the effect of the diluting solvent on the fluorescence response was examined using several common solvents, including DMSO, methanol, ethanol, acetonitrile, and water. Among these, DMSO provided the highest fluorescence intensity, likely due to its favorable solvation properties and compatibility with the formed fluorophore. Consequently, DMSO was selected as the diluting solvent throughout the study.

Method validation

The proposed spectrofluorimetric method was validated in accordance with ICH Q2(R2) guidelines to ensure its reliability for quantitative analysis of IBX. The method demonstrated excellent linearity over the concentration range of 150–2000 ng/mL. The calibration curve exhibited a high coefficient of determination (r^2 = 0.9996), indicating a strong linear correlation between corrected fluorescence intensity and IBX concentration. The regression equation parameters, including a slope of 0.3941 ± 0.0035 (standard error) and an intercept of 48.1977 ± 3.9347 (standard error), further confirmed the sensitivity and consistency of the method (Table 1).

The limits of detection (LOD) and quantification (LOQ) were calculated based on the standard deviation of the response (σ) and the slope (S) of the calibration curve, in accordance with ICH guidelines, using the equations: LOD = 3.3 σ/S and LOQ = 10 σ/S. The calculated LOD and LOQ values were 39.196 ng/mL and 118.774 ng/mL, respectively. These low values indicate the high sensitivity of the proposed spectrofluorimetric method.

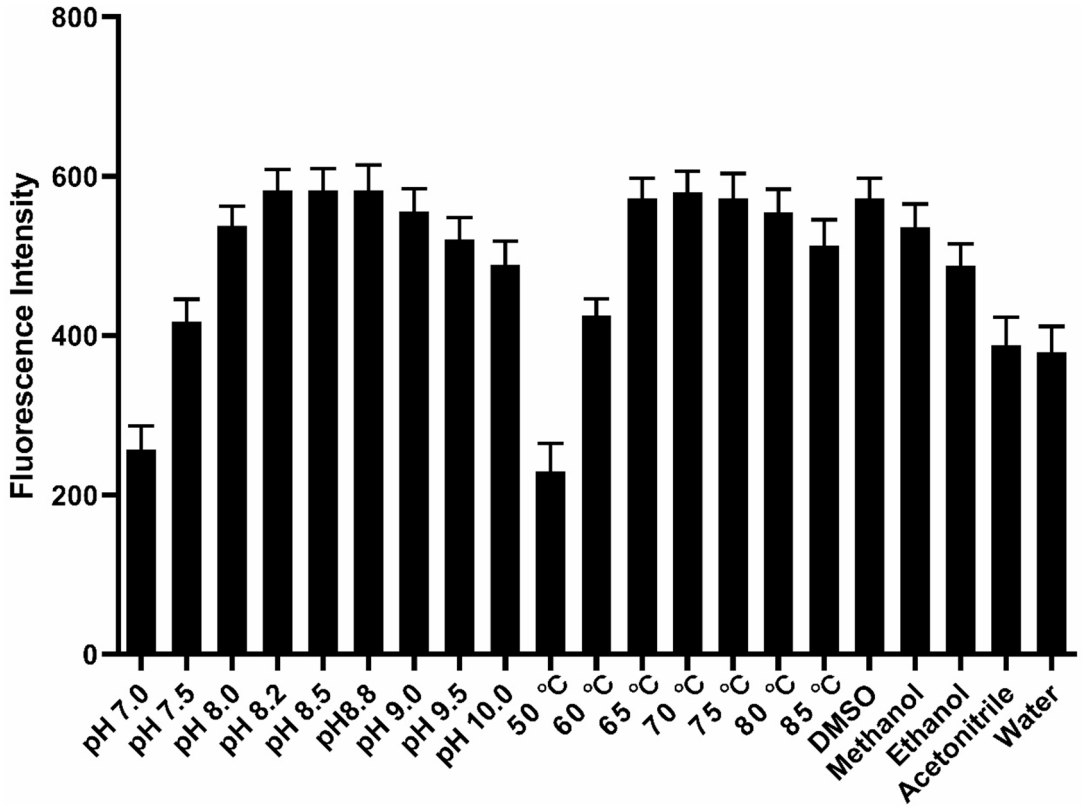


Fig. 4. Effect of pH, temperature, and solvents on the corrected fluorescence intensity of NBD-IBX derivative. Data represent mean ± SD (n = 3).

Parameters	IBX
Emission wavelength (nm)	554
Excitation wavelength (nm)	472
Linearity range (ng/mL)	150–2000
Slope	0.3941
SE of slope	0.0035
Intercept	48.1977
SE of intercept	3.9347
Coefficient of determination (r^2)	0.9996
LOD (ng/mL)	39.196
LOQ (ng/mL)	118.774

Table 1. Regression and validation data for the determination of IBX by the proposed method.

Concentration (ng/mL)	Accuracy (%R*+SD)	Intra-day precision (RSD*)	Inter-day precision (RSD*)
300	99.34 ± 0.845	0.850	0.993
1000	99.72 ± 0.874	0.876	1.016
1600	99.94 ± 0.918	0.919	1.009

Table 2. Accuracy and precision data. *Intra-day: three replicates on the same day; Inter-day: three replicates on three consecutive days ($n = 9$ total).

Method accuracy was assessed via recovery studies at three concentration levels representing low (300 ng/mL), medium (1000 ng/mL), and high (1600 ng/mL) points within the calibration range, each analyzed in triplicate. As shown in Table 2, the percent recovery values were 99.34 ± 0.845, 99.72 ± 0.874, and 99.94 ± 0.918% at 300, 1000, and 1600 ng/mL, respectively, with an overall mean recovery of 99.67%, demonstrating the high accuracy of the method and its suitability for quantitative determination. Precision was evaluated in terms of intra-day and inter-day precision. Intra-day precision was assessed by analyzing the three concentration levels in triplicate within a single analytical session, while inter-day precision was determined by repeating the same measurements on three consecutive days, yielding nine determinations in total for each parameter. The intra-day RSD values were 0.850, 0.876, and 0.919% at 300, 1000, and 1600 ng/mL, respectively, while the inter-day RSD values were 0.993, 1.016, and 1.009% at the same concentration levels. All RSD values were well within the ICH Q2(R2)-accepted limit of 2%, confirming the excellent precision and reproducibility of the proposed method (Table 2).

The stability of the IBX-NBD derivative was evaluated by measuring the corrected fluorescence intensity at 0, 1, 2, 4, 8, and 24 h after derivatization under ambient conditions (25 °C, protected from light). The relative fluorescence intensities at these time points were 100.0, 99.6, 99.3, 98.9, 98.7, and 98.4% of the initial value, respectively, with a maximum decrease not exceeding 2% throughout the entire tested period, confirming the high stability of the formed reaction product and its suitability for routine batch analysis.

The robustness of the method was examined by introducing small, deliberate variations in key experimental parameters, including heating temperature (65–75 °C), heating time (20–30 min), pH (8.3–8.7), and volumes of buffer (0.75–1.25 mL) and NBD-Cl reagent (1.25–1.75 mL). The obtained recovery values ranged between 98.63% and 100.76%, with low RSD values, demonstrating that these minor variations did not significantly affect the analytical performance. These findings confirm the robustness and reliability of the proposed method under slightly varied experimental conditions (Table 3).

To evaluate the specificity of the proposed method, the potential interference from tablet excipients present in Brexafemme® tablets was investigated. Known amounts of butylated hydroxyanisole (BHA), colloidal silicon dioxide, crospovidone, magnesium stearate, mannitol, and microcrystalline cellulose were individually added in excess to a fixed concentration of IBX (1000 ng/mL). The tolerance limit was defined as the maximum amount of each excipient that caused less than ± 2% change in the percent recovery of IBX. As shown in Table 4, all tested excipients were well tolerated at up to 50-fold excess over IBX without causing significant interference, yielding mean recoveries ranging from 98.84% to 100.42%. These results confirm the high specificity of the proposed method and its suitability for direct application to pharmaceutical tablet analysis without interference from formulation components.

Application of the proposed method

The developed method was successfully applied to the determination of IBX in its pharmaceutical dosage form. No interference from common tablet excipients was observed, demonstrating the specificity of the method. The obtained results (Table 5) were in good agreement with the labeled claim and were further validated using the standard addition technique. Moreover, the analytical performance of the developed method was statistically compared with that of a previously reported HPLC method[8] using Student's t-test and F-test at a 95% confidence

Variation	Percent recovery* ± RSD
Optimum conditions	100.37 ± 0.457
Heating temperature	
65 °C	100.76 ± 0.858
75 °C	100.63 ± 0.650
Heating time	
20 min	98.63 ± 0.586
30 min	99.24 ± 1.075
Effect of pH	
pH 8.3	100.07 ± 0.924
pH 8.7	99.07 ± 0.665
Volume of buffer	
0.75 mL	100.40 ± 1.010
1.25 mL	99.88 ± 0.847
Volume of NBD-chloride	
1.25 mL	100.50 ± 0.704
1.75 mL	99.86 ± 1.344

Table 3. Evaluation of robustness of the proposed spectrofluorimetric method. *Average of three determinations.

Excipient	Amount added (µg/mL)	Fold excess over IBX	Recovery (%) ± RSD
Butylated hydroxyanisole (BHA)	10,000	10	99.38 ± 0.74
Colloidal silicon dioxide	50,000	50	100.42 ± 0.91
Crospovidone	50,000	50	98.84 ± 1.12
Magnesium stearate	10,000	10	99.61 ± 0.83
Mannitol	50,000	50	99.05 ± 1.05
Microcrystalline cellulose	50,000	50	100.17 ± 0.96

Table 4. Effect of common tablet excipients on the recovery of IBX by the proposed spectrofluorimetric method ($n = 3$). IBX concentration: 1000 ng/mL. Tolerance limit defined as the maximum excipient amount causing < ± 2% change in IBX recovery.

Sample	Percent recovery ± RSD		t-test (2.306)*	F-test (6.388)*
	Proposed method	Reported method [8]		
Brexafemme ® Tablets**	99.45 ± 0.616	100.15 ± 1.065	1.288	3.031
Standard addition ***	99.85 ± 1.116			

Table 5. Determination of IBX in pharmaceutical formulation using the proposed method and statistical comparison with a reported method. *The values in parenthesis are tabulated values of "t "and "F" at (P = 0.05). **Average of five determinations. *** Average of three determinations.

level. The calculated values (Table 5) were lower than the corresponding critical values, indicating that there is no statistically significant difference between the two methods with respect to accuracy and precision.

Comparative study with reported HPLC method

A comparative evaluation between the proposed spectrofluorimetric method and the previously reported HPLC method[8] was performed (Table 6). The proposed method demonstrated superior sensitivity, with significantly lower limits of detection (0.04 µg/mL) and quantification (0.12 µg/mL) compared to the HPLC method (0.13 µg/mL and 0.38 µg/mL, respectively). Additionally, the proposed method operates over a lower and more suitable concentration range (0.15–2 µg/mL), making it more applicable for trace-level analysis. The practical significance of this sensitivity advantage is further supported by clinical pharmacokinetic data, whereby the reported mean peak plasma concentration (Cmax) of IBX following the approved therapeutic dose of 300 mg twice daily is 435 ng/mL under fasted conditions and 629 ng/mL under fed conditions. Both values fall directly within the working range of the proposed method, whereas the previously reported HPLC method operates at concentrations 30–70 times higher than the expected therapeutic Cmax, confirming its unsuitability for direct bioanalytical application without extensive sample preconcentration. These findings highlight the potential applicability of the

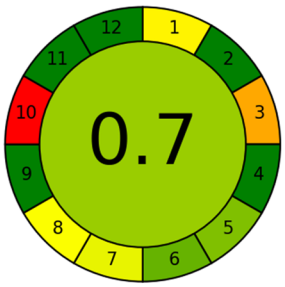
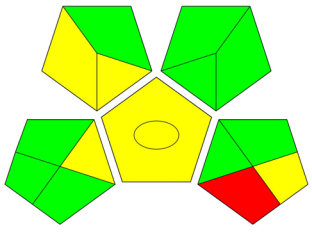
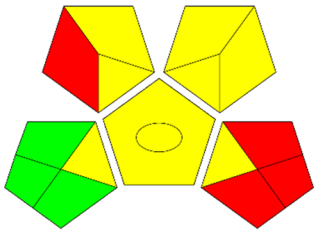
Parameter	Proposed method		Reported method	
Instrument	Spectrofluorimetry		HPLC	
Linearity	0.15–2 µg/mL		15–90 µg/mL	
LOD	0.04 µg/mL		0.13 µg/mL	
LOQ	0.12 µg/mL		0.38 µg/mL	
Solvent	DMSO		DMSO + Acetonitrile	
Analytical eco-scale	Parameters	Penalty points	Parameters	Penalty points
	DMSO	0	DMSO	0
	Borate buffer	2	Acetonitrile	8
	NBD-Cl	1	Phosphate buffer	0
	Energy	0	Energy	1
	Occupational hazards	0	Occupational hazards	0
	Waste	6	Waste	8
	∑ Penalty points	9	∑ Penalty points	17
	Total scores	100-9 = 91	Total scores	100 − 17=83
AGREE				
GAPI				

Table 6. Comparison between the proposed and reported methods.

proposed method for future pharmacokinetic and therapeutic drug monitoring studies following appropriate sample preparation and matrix validation.

From an environmental perspective, the proposed method showed a higher Analytical Eco-scale[14] score (91) compared to the reported HPLC method (83), reflecting reduced use of hazardous solvents and lower waste generation. Furthermore, the avoidance of acetonitrile enhances method safety and sustainability.

Greenness assessment using the AGREE tool[15] further confirmed the environmental advantage of the proposed method, demonstrating a higher overall greenness score compared to the reported HPLC

method, which is mainly attributed to simpler procedure, lower energy consumption, and reduced use of toxic reagents. In addition, GAPI evaluation[16] revealed a more favorable green profile for the proposed method, with fewer red zones and more green segments, indicating lower environmental impact across various analytical steps, particularly in sample preparation and solvent usage.

These advantages, combined with operational simplicity, lower instrumentation cost, simpler sample preparation requirements, and superior greenness profile, confirm the superiority of the proposed method as a more accessible and environmentally sustainable alternative for the determination of IBX in pharmaceutical formulations. It is acknowledged that the derivatization step requires approximately 25 min; however, this is offset by the elimination of mobile phase preparation, column equilibration, and system suitability testing typically associated with chromatographic methods, rendering the overall operational procedure considerably less demanding in terms of instrumentation and technical expertise.

Conclusion

In conclusion, a sensitive, reliable, and environmentally friendly spectrofluorimetric method was successfully developed for the determination of IBX based on derivatization with NBD-Cl. The proposed method effectively overcomes the limitation of weak native fluorescence of IBX, enabling its accurate determination at low concentration levels. Optimization of the experimental conditions ensured maximum fluorescence intensity and high reproducibility. The method exhibited excellent analytical performance, including good linearity, accuracy, precision, sensitivity, and robustness, in full agreement with ICH Q2(R2) validation guidelines. Its successful application to pharmaceutical tablets confirmed its specificity, with no interference from commonly encountered excipients. Compared with conventional chromatographic techniques, the proposed method offers significant advantages including operational simplicity, lower instrumentation cost, reduced solvent consumption, and simpler sample preparation, without the requirement for mobile phase preparation or column equilibration. The superior sensitivity of the method, with a working range of 150–2000 ng/mL, aligns directly with the reported therapeutic plasma concentrations of IBX (Cmax 435–629 ng/mL), supporting its potential future application in bioanalytical and pharmacokinetic studies. In addition, the favorable results obtained from greenness assessment tools emphasize its suitability as a sustainable analytical approach. Future work may focus on adapting and validating the method for pharmacokinetic studies and therapeutic drug monitoring in biological matrices, as well as exploring its applicability to other compounds with similar chemical structures.

Data availability

The datasets used and/or analyzed during the current study are available from the corresponding author on reasonable request.

Received: 28 April 2026; Accepted: 18 June 2026
Published online: 23 June 2026

References

1. Jallow, S. & Govender, N. P. Ibrexafungerp: a first-in-class oral triterpenoid glucan synthase inhibitor. *J. Fungi*. **7**, 163. https://doi.org/10.3390/jof7030163 (2021).
2. Huang, X., Chen, M., Chen, Z. & Zhang, Y. Fungal β-1, 3-glucan synthase: a review of structure, mechanism, and regulation. *FEMS Yeast Res.* **25**, foaf071. https://doi.org/10.1093/femsyr/foaf071 (2025).
3. Lee, A. Ibrexafungerp: first approval. *Drugs* **81**, 1445–1450. https://doi.org/10.1007/s40265-021-01571-5 (2021).
4. Gamal, A. et al. Ibrexafungerp, a novel oral triterpenoid antifungal in development: overview of antifungal activity against Candida glabrata. *Front. Cell. Infect. Microbiol.* **11**, 642358. https://doi.org/10.3389/fcimb.2021.642358 (2021).
5. Sucher, A. J. et al. Ibrexafungerp: a new triterpenoid antifungal. *Am. J. Health-System Pharm.* **79**, 2208–2221. https://doi.org/10.1093/ajhp/zxac256 (2022).
6. Phillips, N. A., Rocktashel, M. & Merjanian, L. Ibrexafungerp for the treatment of vulvovaginal candidiasis: design, development and place in therapy. *Drug Des., Dev. Ther.* 17, 363–367 (2023). (2023). https://doi.org/10.2147/DDDT.S339349
7. Walley, A. H., Dupuis, L. N., Woodahl, E. L. & Killam, S. R. Ibrexafungerp: mechanism of action, clinical, and translational science. *Clin. Transl. Sci.* 18, e70362 (2025). (2025). https://doi.org/10.1111/cts.70362
8. Bindu, M., Sreeja, M., Kusuma, S. & Priya, V. Analytical method development and validation of ibrexafungerp by RP-HPLC method. *World J. Pharm. Pharm. Sci.* **13**, 2142–2152 (2024).
9. Gore, M. (ed) *Spectrophotometry and spectrofluorimetry: a practical approach* Vol. 225 (OUP Oxford, 2000).
10. Mohamed, A. A., Maklad, R. M., Zeid, A. M. & Mostafa, I. M. Complexation-based spectrofluorimetric method for besifloxacin determination: Combined experimental and computational insights. *Anal. Chim. Acta*. **1372**, 344446. https://doi.org/10.1016/j.aca.2025.344446 (2025).
11. Rudolf Seitz, W. & Frei, R. W. Fluorescence derivatization. Crit. Rev. *Anal. Chem.* **8**, 367–405. https://doi.org/10.1080/10408348008542715 (1980).
12. Vashistha, V. K., Bala, R. & Pullabhotla, V. S. R. Derivatizing agents for spectrophotometric and spectrofluorimetric determination of pharmaceuticals: a review. *J. Taibah Univ. Sci.* **17**, 2206363. https://doi.org/10.1080/16583655.2023.2206363 (2023).
13. Alaqel, S. I. et al. Application of a nucleophilic substitution reaction for spectrofluorimetric determination of aripiprazole in pharmaceutical dosage form and plasma matrix; greenness assessment. *Spectrochim Acta Part. A*. **303**, 123238. https://doi.org/10.1016/j.saa.2023.123238 (2023).
14. Gałuszka, A., Migaszewski, Z. M., Konieczka, P. & Namieśnik, J. Analytical Eco-Scale for assessing the greenness of analytical procedures. *TrAC Trends Anal. Chem.* **37**, 61–72. https://doi.org/10.1016/j.trac.2012.03.013 (2012).
15. Pena-Pereira, F., Wojnowski, W. & Tobiszewski, M. AGREE—Analytical GREEnness metric approach and software. *Anal. chem.* **92**, 10076–10082. https://doi.org/10.1021/acs.analchem.0c01887 (2020).
16. Płotka-Wasylka, J. A new tool for the evaluation of the analytical procedure: Green Analytical Procedure Index. *Talanta* **181**, 204–209. https://doi.org/10.1016/j.talanta.2018.01.013 (2018).

Acknowledgements
The authors extend their appreciation for Princess Nourah bint Abdulrahman University Researchers Supporting Project number (PNURSP2026R917), Princess Nourah bint Abdulrahman University, Riyadh, Saudi Arabia.

Author contributions
OA: Methodology, Writing original draft. MS; Validation, Writing original draft. All authors reviewed the manuscript.

Funding
This research was funded by Princess Nourah bint Abdulrahman University Researchers Supporting Project number (PNURSP2026R917), Princess Nourah bint Abdulrahman University, Riyadh, Saudi Arabia.

Declarations

Competing interests
The authors declare no competing interests.

Additional information
Correspondence and requests for materials should be addressed to M.S.

Reprints and permissions information is available at www.nature.com/reprints.

Publisher's note Springer Nature remains neutral with regard to jurisdictional claims in published maps and institutional affiliations.

Open Access This article is licensed under a Creative Commons Attribution-NonCommercial-NoDerivatives 4.0 International License, which permits any non-commercial use, sharing, distribution and reproduction in any medium or format, as long as you give appropriate credit to the original author(s) and the source, provide a link to the Creative Commons licence, and indicate if you modified the licensed material. You do not have permission under this licence to share adapted material derived from this article or parts of it. The images or other third party material in this article are included in the article's Creative Commons licence, unless indicated otherwise in a credit line to the material. If material is not included in the article's Creative Commons licence and your intended use is not permitted by statutory regulation or exceeds the permitted use, you will need to obtain permission directly from the copyright holder. To view a copy of this licence, visit http://creativecommons.org/licenses/by-nc-nd/4.0/.

© The Author(s) 2026